

POWER INPUT

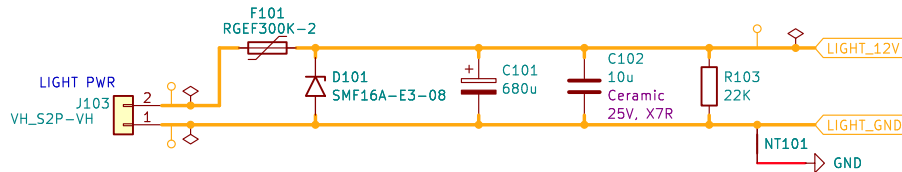
Input must be 5 to 16 V

Circuit protection

- Internal: 3 A polyfuse and 16 V TVS
- External: 5 A fuse recommended

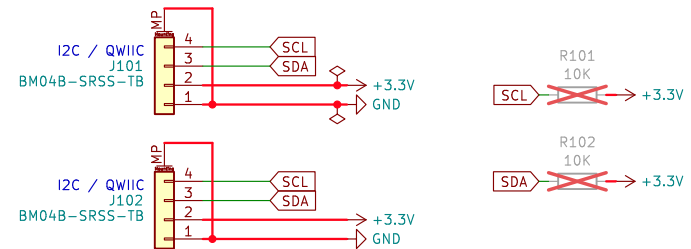
Bulk capacitor sizing assumptions:

- Switching maximum 3 A at minimum 400 Hz PWM
- $Q_{bulk} = 3 \text{ A} \cdot (1 / 400 \text{ s}) = 0.0075 \text{ C}$
- $C_{bulk} = Q_{bulk} / 12 \text{ V} = 625 \text{ uF}$



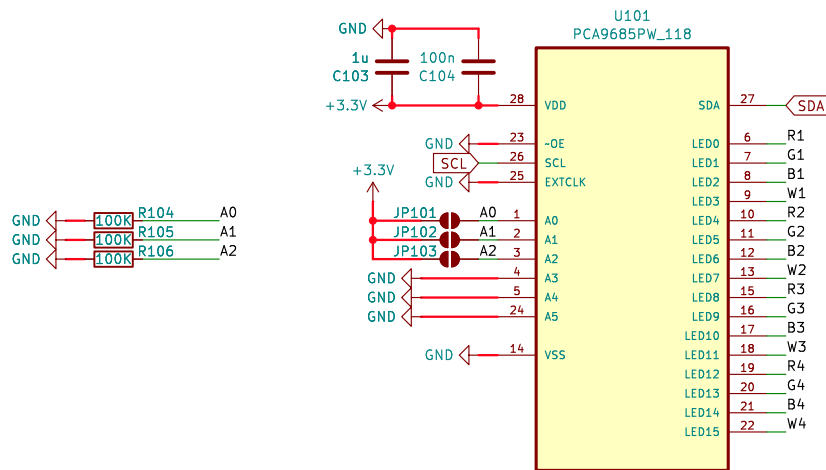
QWIIC INTERFACE

Pull-ups can be omitted assuming the host already has them
Use both ports to form a daisy chain

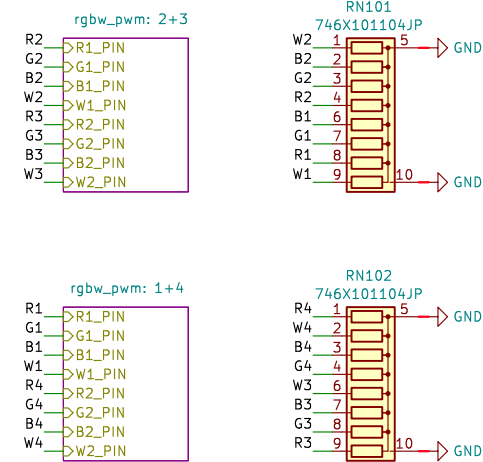


I2C PWM DRIVER

Can be programmed to one of 8 unique addresses



LOW-SIDE LED DRIVERS



- 4x4 channel RGB/W LED driver
- Input voltage range: 12 V to 14.5 V
 - Output max rating: 3 A, 36 W
 - LED PWM frequency: 1000 Hz
 - Fixture max power: 0.5 A spots (6 W), 3 A strips (36 W)
 - Fixture typical usage: 25% of maximum

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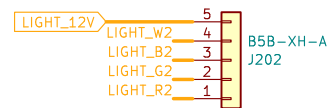
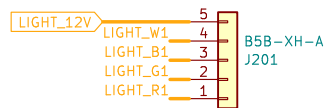
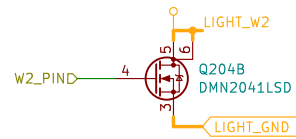
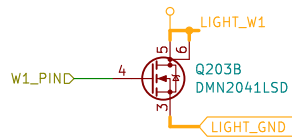
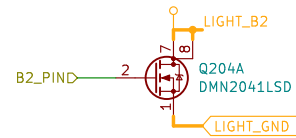
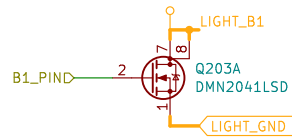
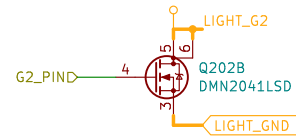
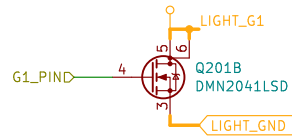
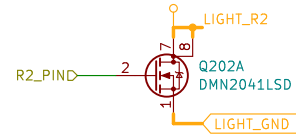
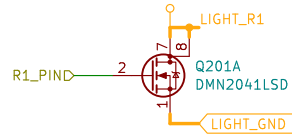
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Rev:

Id: 1/3

LOW-SIDE N-CHANNEL MOSFET SWITCHES

DMN2014LSD has $R_{DSon} < 40 \text{ m}\Omega$ at $V_{gs} = 3.3 \text{ V}$, handles $> 4 \text{ A}$ at 25°C ambient
Outputs arranged in 2 groups of 4
Includes a common anode connection for LED lamps



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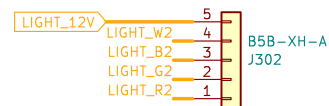
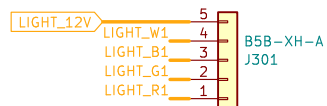
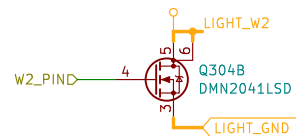
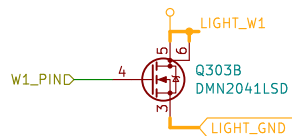
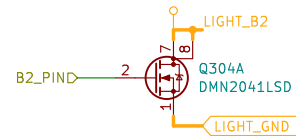
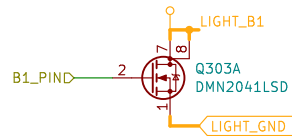
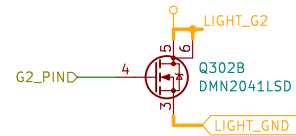
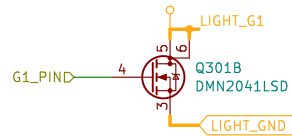
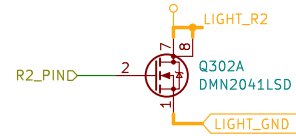
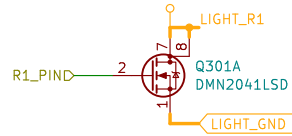
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